PHASE 1 DELIVERY HANDOFF

SAQR Production-Ready Handoff Document

Formal handoff package for the delivery engineering team. This document consolidates the repo truth, delivery obligations, technical boundaries, validation path, and Phase 1 non-negotiables into one presentation-ready handover artifact.

AUDIENCE

Internal Delivery Team

STATUS

Delivery-Handoff Ready

PHASE

Phase 1 Complete

DATE

2026-04-07

Prepared From Repo State

SAQR merged Phase 1 handoff baseline on GitHub main.

Document Intent

Equip delivery to continue without missing prerequisites, constraints, or validation steps.

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EXECUTIVE SUMMARY

1. Executive Summary

SAQR Phase 1 is complete in repository scope. The platform has been moved from a pre-MVP demo state to a delivery-handoff-ready package with separated demo and production-ready runtime behavior, documented service contracts, hardened startup validation, provider-aware seams, deployment packaging, acceptance fixtures, and a complete validation path.

Formal Verdict

SAQR is ready for delivery-team implementation handoff. SAQR is not ready for direct client production cutover until delivery-owned infrastructure wiring and real-environment pilot validation are completed.

REPO SCOPE

100%

All tracked Phase 1 items are completed in the repository.

DELIVERY DEPENDENCIES

6

Major external workstreams remain outside repo scope.

CORE POSITION

Read-Only

SAQR must not write back to client source systems.

What delivery must understand immediately

- The current demo experience is intentionally preserved and must continue to exist as a separate path.
- The UI/UX baseline is frozen. Delivery must not redesign or reflow the interface in Phase 1.
- The repo ships interface-ready algorithms, not live client infrastructure.
- Current live regulatory-source scope is centered on SAMA and SDAIA. Anything beyond that is additional delivery work.
- Phase 2 features are not part of this handoff and must not be implied as included.

PROJECT OVERVIEW

2. Project Overview

2.1 Platform purpose

SAQR is a read-only compliance interception platform. It observes regulatory change and client operating evidence, detects non-compliance against codified rules, preserves evidence, and exposes that evidence through the SAQR API and Shield UI.

The Phase 1 target was not enterprise feature expansion. It was to turn the product core into a handoff-ready baseline that the delivery team can safely wire into real environments.

2.2 Non-negotiables that were preserved

- Demo environment retained in full.
- UI/UX preserved with regression guardrails.
- No assumed live DB or third-party environment in repo.
- Only free tooling used in the build path.
- Documentation produced alongside implementation.

PROJECT FACT	PHASE 1 POSITION	DELIVERY INTERPRETATION
Product state	Repo complete	Delivery can proceed without first re-architecting the core.
Client cutover	Not included	Requires environment wiring, security controls, and pilot validation.
Demo experience	Preserved	Do not repurpose or break the demo path.
UI redesign	Prohibited	Only agreed and approved exceptions may touch the visual baseline.

READINESS POSITION

3. Phase 1 Readiness Position

COMPLETED IN REPO	Demo and production-ready runtime separation, auth scaffolding, config safety, observability, service contracts, provider boundaries, rule modularization, deployment packaging, acceptance fixtures, CI quality gates, release-readiness automation, and the final handoff package.
DELIVERY-OWNED	Identity provider wiring, shadow database provisioning, Kafka and Kafka Connect / Debezium, VMS connectivity, infrastructure security controls, authority expansion, and pilot validation in the target environment.
EXPLICITLY EXCLUDED	LogicGate-style no-code workflow engine, Archer-style multi-entity and sovereign expansion, and any claim that the repo alone is sufficient for direct client go-live.

Do Not Misstate Readiness

Phase 1 completion means the repository is handoff-ready for delivery implementation. It does not mean SAQR is turnkey production-live. Any communication to the client must preserve that distinction.

ARCHITECTURE

4. Solution and Architecture Overview

The production-ready SAQR topology is intentionally split into observation, interpretation, evidence, and presentation layers. The system remains read-only against client source systems.

COMPONENT	PRIMARY ROLE	RUNTIME ENTRY / KEY ARTIFACT	DELIVERY NOTE
Shield UI	Frozen Phase 1 interface shell for dashboard, evidence visibility, and operator workflow.	`apps/shield-ui`	Use unchanged UI; configure runtime injection only.
SAQR API	Read API for dashboard metrics, staging visibility, evidence, and health metadata.	`apps/api/src/index.js`	Wire real bearer token issuance to documented JWT contract.
Evidence Vault	CDC processing, evidence sealing, rule evaluation, and Merkle logging.	`services/evidence-vault/src/index.js`	Needs Kafka and shadow DB.
Sentinel Scrapers	Regulatory-source collection and staging.	`services/sentinel-scrapers/src/index.js`	Live scope currently centered on SAMA and SDAIA.
NLP Interpreter	Circular parsing, obligation extraction, and drift detection.	`services/nlp-interpreter/src/index.js`	Interface-ready and test-backed; still delivery-wired in real environments.
CV Watchman	VMS scan-cycle processing, detection, and evidence creation.	`services/cv-watchman/src/index.js`	Requires delivery-side VMS provider connection and calibration.
CDC Connector Artifacts	Debezium connector registration and source DB contract.	`services/cdc-connector`	Artifact set, not a runnable Node service.

4.1 Runtime modes

RUNTIME	INTENDED USE	BEHAVIOR
`demo`	Client demo continuity	Allows embedded demo data, simulation helpers, and silent fallback paths.
`production-ready`	Delivery handoff baseline	Rejects demo-only execution modes and enforces fail-fast config rules.

4.2 Core architectural rules

- SAQR remains read-only to client source systems.
- Runtime enforcement is centralized through shared profile utilities.
- Config validation is centralized through shared service-config utilities.
- Provider boundaries are explicit so delivery can swap infrastructure adapters without rewriting core logic.
- Observability is structured and shared across services.

ARTIFACT INVENTORY

5. Repository Artifact Map

The following artifacts are the canonical handoff package entrypoints and supporting references the delivery team should use first.

CATEGORY	KEY ARTIFACTS	PURPOSE

Primary handoff entrypoints	`docs/handoff/README.md`
	`docs/handoff/SAQR_Phase1_Handoff_Summary.md`
	`docs/handoff/SAQR_Phase1_Delivery_Worklist.md`
	`docs/handoff/SAQR_Phase1_Validation_Guide.md`
		Canonical starting point for delivery pickup.

Architecture and runbooks	`docs/architecture/SAQR_Phase1_Architecture_Overview.md`
	`docs/runbooks/SAQR_Phase1_Operations_Runbook.md`
	`docs/runbooks/SAQR_Phase1_Dependency_Incident_Runbook.md`
		Operating model, dependency awareness, and troubleshooting guidance.

Contracts	`docs/contracts/saqr-service-contracts.md`
	`docs/contracts/saqr-api.openapi.yaml`
	`docs/contracts/saqr-execution-sequences.yaml`
	`docs/contracts/saqr-integration-adapters.md`
		Delivery implementation contract and system boundary definition.

Deployment and data	`infra/docker-compose.production.yml`
	`infra/k8s/base/*`
	`infra/migrations/*`
	`docs/contracts/saqr-data-dictionary.md`
		Packaging templates, migration baseline, and schema governance.

Validation	`package.json`
	`tools/ci/run-phase1-quality-gates.js`
	`tools/ci/run-phase1-release-readiness.js`
	`tools/ci/verify-phase1-handoff-package.js`
		Free-toolchain validation and release-readiness flow.

RESPONSIBILITIES

6. Delivery Workstreams and Responsibilities

WORKSTREAM	REPO STATUS	DELIVERY TEAM MUST DO	COMPLETION EVIDENCE
Identity and API access	Delivery action	Wire a real token issuer or IdP, issue compatible JWTs, validate access from frozen UI.	Authenticated API requests succeed with delivery-issued tokens.
Shadow database and schema	Delivery action	Provision target PostgreSQL, apply migrations, validate least-privilege access for services.	Startup validation passes and shadow tables are queryable.
CDC and messaging	Delivery action	Provision Kafka and Kafka Connect / Debezium, register connector artifacts, validate event flow.	CDC topics populate and Evidence Vault consumes correctly.
Regulatory sources	Delivery action	Validate live SAMA and SDAIA scraping in target network conditions. Add more authorities only if needed.	Stable live staging ingestion with documented selector behavior.
CV and VMS	Delivery action	Connect the real VMS provider, validate scan-cycle behavior, calibrate pilot detection behavior.	Real camera connectivity and validated evidence persistence.
Platform deployment	Delivery action	Replace placeholders, tailor compose or Kubernetes packaging, configure ingress, TLS, registry, and network controls.	Target-environment deployment plan and secure runtime manifest.
Demo preservation	Complete in repo	Keep the demo environment separate from the production-ready deployment path.	Demo path remains runnable and unchanged.
Phase 2 feature set	Excluded	Do not scope or imply workflow-engine or multi-entity features as part of this handoff.	Phase 2 tracked independently.

Recommended Delivery Order

1. Provision shadow DB and schema. 2. Wire auth and API access. 3. Establish Kafka and CDC flow. 4. Validate Sentinel and NLP. 5. Connect CV to real VMS. 6. Apply target deployment controls. 7. Run full validation and pilot-readiness signoff.

CONFIGURATION BASELINE

7. Environment and Configuration Baseline

Delivery must preserve the two-track configuration model and respect the production-ready fail-fast rules. No real secrets are stored in the repository.

AREA	VARIABLES / CONTROLS	DELIVERY GUIDANCE
Global runtime	`SAQR_RUNTIME_MODE`, `SAQR_LOG_LEVEL`, `SAQR_AUDIT_LOG_ENABLED`	Use `production-ready` behavior for delivery environments and keep demo isolated.
API	`API_HOST`, `API_PORT`, `AUTH_ENABLED`, `JWT_SECRET`, `AUTH_JWT_ISSUER`, `AUTH_JWT_AUDIENCE`	Do not disable auth in the production-ready track.
Shared DB	`SHADOW_DB_HOST`, `SHADOW_DB_PORT`, `SHADOW_DB_NAME`, `SHADOW_DB_USER`, `SHADOW_DB_PASSWORD`	Placeholder values are rejected in production-ready mode.
Evidence Vault	`KAFKA_BOOTSTRAP_SERVERS`, `EVIDENCE_VAULT_TOPICS`, `NTP_SERVER`	Kafka and NTP validation are part of delivery readiness.
Sentinel	`SENTINEL_MODE`, `SENTINEL_CRON`, `SENTINEL_BROWSER_HEADLESS`	`demo` mode is not acceptable in the production-ready track.
NLP	`NLP_BOOT_MODE`, `NLP_VALIDATE_DB_ON_STARTUP`	`demo-ingest` is rejected in production-ready mode.
CV Watchman	`VMS_TYPE`, `CV_VMS_HOST`, `CV_VMS_USERNAME`, `CV_VMS_PASSWORD`	`demo` VMS is not acceptable in the production-ready track.
UI runtime injection	`SAQR_UI_API_BASE`, `SAQR_UI_AUTH_ENABLED`, `SAQR_UI_TOKEN_STORAGE_KEY`	UI configuration only; no visual redesign required.

Source templates

- `.env.demo.example` keeps the current demo baseline.
- `.env.production.example` is the delivery handoff placeholder template.
- `shared/runtime-profile.js` controls demo versus production-ready behavior.

- `shared/service-config.js` centralizes placeholder rejection and startup validation.

Configuration hygiene rules

- Never commit real client secrets into the repository.
- Do not bypass runtime validation to force a service up.
- Keep config changes synchronized with templates, docs, and contracts.
- Treat localhost warnings in production-ready mode as delivery action items.

EXTERNAL DEPENDENCIES

8. External Prerequisites

Critical Dependency Notice

The repository does not provision the client environment. Delivery must bring the external dependencies below before any production-ready deployment claim is made.

PREREQUISITE	OWNER	REQUIRED FOR	IF MISSING
Shadow PostgreSQL	Client or delivery team	API, Evidence Vault, NLP, CV, Sentinel	Core persistence and visibility paths do not function.
Kafka and Kafka Connect / Debezium	Client or delivery team	CDC connector and Evidence Vault	CDC evidence path is not live.
Identity provider or token issuer	Delivery team	Authenticated API access from Shield UI	Production-ready auth contract cannot be exercised.
VMS endpoint and credentials	Client or delivery team	CV Watchman	Real camera scan flow remains unvalidated.
Ingress, DNS, TLS, storage, and cluster controls	Delivery team	Compose / Kubernetes deployment	No secure client deployment posture exists.

VALIDATION

9. Validation and Signoff Path

STEP	COMMAND	EXPECTED SIGNAL
Workspace bootstrap	`npm run phase1:bootstrap`	`workspace bootstrap complete`
Handoff package verification	`npm run phase1:handoff:verify`	PASS line confirming manifest and references are valid
Quality gates	`npm run phase1:quality`	All listed gates report PASSED
Release-readiness	`npm run phase1:release:verify`	`release-readiness verification complete`
Container build smoke test	`npm run phase1:release:verify -- --build-containers`	Production-ready images build successfully
UI freeze verification	`npm --prefix apps/shield-ui run ui:baseline:check`	UI baseline check passes for all tracked files

Repo package signoff

Bootstrap, handoff verification, quality gates, and release-readiness must all pass.

Delivery-environment signoff

DB, Kafka, IdP, VMS, and platform prerequisites must be wired and validated in target conditions.

Pilot-readiness signoff

Real-environment validation must be complete before any production-live claim is made.

RISKS AND LIMITS

10. Risks, Limits, and Explicit Exclusions

Current limits that remain true

- No real client DB, Kafka, IdP, or VMS is provisioned in-repo.
- Sentinel live implementation is currently centered on SAMA and SDAIA.
- NLP and CV are interface-ready and test-backed, but pilot fit must still be validated in the target environment.
- Kubernetes manifests are base templates, not final client manifests.

Items that are explicitly out of scope for Phase 1

- LogicGate-style no-code workflow engine.
- Archer-style multi-entity model.
- Cross-entity reporting and sovereign topology expansion beyond current packaging.
- Any promise of turnkey client go-live from repo contents alone.

Delivery Messaging Rule

Describe this handoff as a production-ready implementation package for the delivery team. Do not describe it as a client-ready production deployment until delivery-owned prerequisites and pilot evidence exist.

APPENDICES

11. Appendices

11.1 Key file paths

PURPOSE	PATH
Primary handoff package	`docs/handoff/README.md`
Service contracts	`docs/contracts/saqr-service-contracts.md`
Architecture overview	`docs/architecture/SAQR_Phase1_Architecture_Overview.md`
Deployment blueprint	`docs/SAQR_Phase1_Deployment_Blueprint.md`
Configuration guide	`docs/guides/SAQR_Phase1_Configuration_Guide.md`
Release checklist	`docs/checklists/SAQR_Phase1_Release_Checklist.md`

11.2 Delivery handoff statement

Final Handoff Statement

Delivery may now take ownership of the SAQR Phase 1 package, wire the client-specific infrastructure, execute the documented validation path, and continue into pilot deployment work without first reworking the core architecture or changing the user experience.

End of document.